

Jaydeep Jitendra Borkar

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🌐 <https://github.com/jaydeepborkar>

EDUCATION & RESEARCH EXPERIENCE

Northeastern University

Sept. 2021 - March 2027 (Expected)

PhD in Computer Sciences

Advisor: Prof. David A. Smith

Thesis committee: David A. Smith, Terra Blevins, Tianshi Li, Niloofar Miresghallah.

Meta Superintelligence Labs

June '26 - Oct. '26

Research Scientist Intern (Safety Evals Team), New York

Research on synthetic classifiers for LLM safety monitoring

Manager: Cyrus Nikolaidis

Meta Superintelligence Labs

June '25 - Nov '25

Visiting Researcher (Pretraining Trust Team), New York

Research on LLM memorization & privacy

Manager: Archi Mitra

Mentors: Zheng Xu, Diego Garcia-Olano, Karan Chadha, Niloofar Miresghallah

MIT-IBM Watson AI Lab

2020 - 2021

External Research Student (remote)

Research on adversarial machine learning.

Advisor: Dr. Pin-Yu Chen

University of Pune

2016 - 2020

Bachelor's degree in Computer Engineering

Research Interests: I'm broadly interested in studying safety in language models (e.g., training data extraction/memorization) and evaluating models for misalignment. I'm also interested in understanding the influence of pretraining data on various model capabilities and methods for curating high-quality synthetic datasets. My memorization research has been cited in a **U.S. congressional testimony** on AI privacy.

PAPERS & RESEARCH PROJECTS

Books, Copyright, and the Future of Foundation Models: An Empirical Study of Fiction in Pre-training

Jaydeep Borkar*, Mohammad Aflah Khan*, Krishna P. Gummadi, David A. Smith, Abhilasha Ravichander

*Equal contribution.

under review at ICLR 2027

Predicting Memorization Before Fine-Tuning Language Models

Jaydeep Borkar, Niloofar Miresghallah, David A. Smith

under review at EACL 2027

Memorization Dynamics in Knowledge Distillation for Language Models

Jaydeep Borkar, Karan Chadha, Niloofar Miresghallah, Yuchen Zhang, Irina-Elina Veliche, Archi Mitra, David A. Smith, Zheng Xu, Diego Garcia-Olano

Conference on Language Modeling (COLM) 2026

Privacy Ripple Effects from Adding or Removing Personal Information in Language Model Training

Jaydeep Borkar, Matthew Jagielski, Katherine Lee, Niloofar Miresghallah, David A. Smith, Christopher A. Choquette-Choo

Association for Computational Linguistics (ACL) 2025

Recite, Reconstruct, Recollect: Memorization in LMs as a Multifaceted Phenomenon

USVSN Sai Prashanth, Alvin Deng, Kyle O'Brien, Jyothir S V, Mohammad Aflah Khan, **Jaydeep Borkar**, Christopher A. Choquette-Choo, Jacob Ray Fuehne, Stella Biderman, Tracy Ke, Katherine Lee, Naomi Saphra
International Conference on Learning Representations (ICLR) 2025

What can we learn from Data Leakage and Unlearning for Law?

Jaydeep Borkar

Generative AI and Law (GenLaw) workshop, ICML 2023

<https://genlaw.github.io/CameraReady/12.pdf>

Mind the gap: Analyzing lacunae with transformer-based transcription

Jaydeep Borkar and David A. Smith

Workshop on Computational Paleography, ICDAR 2024

<https://arxiv.org/abs/2407.00250>

Simple Transparent Adversarial Examples

Jaydeep Borkar and Pin-Yu Chen

Workshop on Security and Safety in Machine Learning Systems, ICLR 2021

<https://aisecure-workshop.github.io/aml-iclr2021/papers/48.pdf>

TALKS

Meta Mechanistic Interpretability Group: talk on memorization during distillation

Google ML Privacy Seminar: talk on memorization and privacy.

Northeastern CS 5100: invited guest lecture on privacy and security in LLMs.

ORGANIZING

Trustworthy ML Initiative

Co-organizer of the Trustworthy ML Initiative along with Prof. Hima Lakkaraju (Harvard), Sara Hooker (Cohere for AI), Dr. Sarah Tan (Salesforce AI), Dr. Subho Majumdar (Vijil), Chhavi Yadav (UC San Diego), Dr. Chirag Agarwal (Harvard), Prof. Haohan Wang (UIUC), and Marta Lemnarczyk (Hasso-Plattner-Institut).

COURSES

Machine Learning CS 6140, Natural Language Processing CS 6120, Deep Learning CS 7150, Machine Learning Security and Privacy CY 7790, Theory and Methods in Human-Computer Interaction CS 7340, AI as an Archival Science CS 7180.

TEACHING EXPERIENCE

Natural Language Processing CS 6120 - TA	Summer 2024
Foundations of AI CS 5100- TA	Spring 2024
Foundations of Data Science DS 3000 - TA	Fall 2023
Product Development for Large Language Models CS 7180 - TA	Summer 2023
Introduction to Computer Science Research CS 3950 and CS 4950 - TA	Spring 2023
Introduction to Machine Learning and Data Mining DA 5030 - TA	Summer and Fall 2022